Okan Berhoğlu

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Profile

Robotics Software Engineer experienced in desktop and industrial software development, machine data acquisition, device communication, and engineering systems. Skilled in communication protocols, database integration, deployment, validation, and technical documentation. Background in Mechanical Engineering and ongoing M.S. studies in Robotics.

Experience

Middle East Technical University — Ankara, Turkey

Researcher — October 2025 - Present

- Conduct research on mobile robotic systems, including kinematic and dynamic modeling, simulation, sensor integration, and autonomous control algorithm development.
- Contribute to a research project investigating human rhythmic behavior. Within the project, I developed an experimental setup using a 3-axis haptic device, collected data from predefined rhythmic motions, and conducted system identification and behavior modeling studies.

Teknopar Industrial Automation — Ankara, Turkey

Software Engineer — February 2024 – October 2025, 1 year 8 months

- Designed the software architecture and developed a Windows service-based data acquisition application for CNC controllers. Integrated TCP/UDP-based industrial communication protocols to collect real-time machine data, standardized the collected data into an MTConnect-like format, and actively contributed to the software's licensing, deployment, and commissioning processes.
- Developed an OSGi-based application with a Swing user interface for military truck systems, enabling operators to monitor onboard power units and control the vehicle's electrical system. Integrated industrial communication protocols such as EtherNet/IP and Modbus TCP for device communication and data collection.
- Actively contributed to the proposal writing of a TÜBİTAK-funded project focused on developing the system model, autonomy stack, and digital twin of a mobile robot. Within the project, worked on AI-based environmental perception and autonomous navigation using ROS2, 3D LiDAR, and depth camera sensors.

Candidate Engineer — December 2022 – February 2024, 1 year 3 months

- Supported R&D activities focused on collecting data from CNC machines and industrial systems.
- Contributed to the modeling of an electromechanical system using Simulink and Simscape.
- Analyzed experimental data from hardware prototypes and supported grey-box system identification and parameter estimation studies.
- Contributed to the integration of the developed system model into IBM Rhapsody, gaining practical experience in model integration and systems engineering workflows.

Intern — May 2022 – December 2022, 8 months

- Worked on data acquisition methods from CNC machines.

Anova R&D Technologies — Ankara, Turkey

Intern — May 2021 – August 2021, 4 months

- Gained hands-on experience in machining, quality control, and finishing processes, and contributed to mold and fixture design activities.

Education

Middle East Technical University — Ankara, Turkey
Master of Science, Robotics — 2024 – 2026

TOBB Economics and Technology University — Ankara, Turkey
Bachelor of Science, Mechanical Engineering — 2018 – 2023

Atatürk University — Remote
Associate degree, Computer Programming, Second University Program — 2021 – 2023

Skills

Software: Java, C#, Python, C++, REST API, Git, Linux, CUDA, Unity, ROS/ROS2, MATLAB, Simulink, Simscape, SQL, InfluxDB
Industrial Communication: TCP/IP, UDP, EtherNet/IP, Modbus TCP, CAN, RS-485, industrial machine communication
Language: English - Upper-Intermediate, TOEFL iBT 2026 score: 5/6 (equivalent to 100/120 in old score system)